

Matthew Garcia

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Portfolio Website: www.matthewgarcia.dev

Education

Northeastern University | Boston, MA

December 2026

Bachelor of Science in Electrical and Computer Engineering, Minor in Mathematics

GPA: 3.72

Relevant Coursework: Cornerstone of Engineering, Computing Fundamentals, Fundamentals of Networks, Embedded Design, Circuits & Signals, Engineering Algorithms, Linear Systems, Digital Design & Computer Organization, Database Design

Activities: IEEE (President), AI NU (Vice President), SHPE (Treasurer), Undergraduate Research & Fellowships Ambassador

Awards: NU-FIRST Robotics Scholarship, Honors Program, Dean's List, PEAK Summit Award, AJC Merit Research Scholarship

Technical Skills

Programming Languages: Python, C, C++, Matlab, Java, Kotlin, TypeScript, SQL, SystemVerilog, RISC-V

Frameworks & Tools: PyTorch, TensorFlow, Flask, PostgreSQL, Processing, Wireshark

Hardware: Raspberry Pi, Arduino, Function Generator, Oscilloscope, Circuit Design, Soldering, 3D Printing, Laser Cutting

Experience

ACT Laboratory

June 2026 – Present

Undergraduate Researcher

Boston, MA

- Enabled control of emergent structural patterns during image generation through extracting **semantic segmentation masks** from cross and self-attention maps from **transformer** within **pre-trained latent diffusion model** with **85% accuracy**
- Applied **inpainting** to blend masks into interpolated latents at defined timesteps, allowing changes within diffusion process

SINRG Laboratory

January 2025 – June 2026

AJC Merit Research Scholar

Boston, MA

- Developed Android application in **Kotlin** integrating multiple STT/LLM/TTS models using **OpenAI and Deepgram APIs**
- Integrated application with Meta Ray-Bans in order for multi-modal AI system to be compatible with images and audio taken with glasses hardware, leading to an **85% reduction in latency** for all tasks compared to other open-source glasses
- Tested latency and throughput of voice/image transmission in glasses, publishing results in paper at **MobiCom 2026**

ZOR!

May 2025 – June 2026

Founding Engineer

Boston, MA

- Led development of machine learning pipelines to extract clinically relevant insights from user input and biosensor data in iOS seizure monitoring app for startup, deployed via **Python** based Azure web service with **100+ current sign ups**
- Created variety of insights, including correlations to find the relationship between health data and seizure frequency, **random forest generators** to find feature importance, and **LSTM models** to extrapolate trends and detect anomalies

National Energy Technology Laboratory

June 2024 – August 2024

Research Associate

Pittsburgh, PA

- Characterized simulated acoustic signals using machine learning to develop early kick detection algorithm in offshore wells
- Incorporated **Fourier transforms and principal component analysis** to enhance feature selection and reduce dimensionality
- Implemented and optimized machine learning models, including **random forest generators, gradient boosting**, and **multilayer perceptron** for classification tasks, increasing **test accuracy by 30%** after hyperparameter tuning

Mayflower Communications Company

January 2024 – May 2024

Software Engineer

Bedford, MA

- Developed **C** software for implementation of digital signal processing algorithms in order to find directions of arrival of meaconing spoofers and generate weights that can null them, allowing planes to use GPS without interference from spoofers
- Implemented **residual neural network (RNN)** converted from **Python to C** to perform automatic modulation classification on signals being received from jammer for five different classes, achieving **88% accuracy rate**

Projects

Research Paper Extraction Tool

github.com/mattgar417/AI-Research-Tool

- Developed tool in **Python** using **vision transformers** to identify and extract tabular data from scientific paper images
- Integrated front-end interface in **HTML** for users to upload papers and view analyzed results

Smart City Resource Allocation

github.com/mattgar417/Smart-City

- Designed and implemented a resource allocation system for smart cities using **linear regression and Q-learning in C++**, optimizing resource flow and allocation for various sectors and achieving efficient system performance and time complexity
- Processed CSV-based input data for resource characteristics and enabled real-time user inputs for enhanced usability

Mecha Mayhem | Project Lead

github.com/mattgar417/Mecha-Mayhem

- Led team of four students to design and develop educational game for Boston Children's Museum that had players assemble laser-cut mecha components onto photoresistors to answer robotics related questions, garnering **95% satisfaction rate**
- Utilized **Arduino and Processing** software to connect electrical components with interface displayed on monitor screen